Homework 20.11.26 – Exercise 40

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Exercise 40 (τ-nice sampling)

Let $S \subseteq \{1, \dots, n\}$ be chosen uniformly at random among all subsets of size τ, where $1 \leq \tau \leq n$. Prove that

$$p_i := \mathbb{P}(i \in S) = \frac{\tau}{n} \qquad \forall i = 1, \dots, n.$$

Proof. Since S is chosen uniformly, p_i is the number of size τ subsets that contain i divided by the number of all size τ subsets. The number of all size τ subsets $S \subseteq \{1, \dots, n\}$ is

$$\binom{n}{\tau} = \frac{n!}{\tau!\,(n-\tau)!}.$$

Now we compute how many of them contain i. Such a subset contains i and $\tau - 1$ other elements, which can be chosen arbitrarily from the remaining $n - 1$ elements, so there are

$$\binom{n-1}{\tau-1} = \frac{(n-1)!}{(\tau-1)!\,(n-\tau)!}$$

of them. Dividing this number by $\binom{n}{\tau}$, the factor $(n-\tau)!$ cancels, and using $n! = n\,(n-1)!$ and $\tau! = \tau\,(\tau-1)!$ we get

$$p_i = \frac{\binom{n-1}{\tau-1}}{\binom{n}{\tau}} = \frac{(n-1)!}{(\tau-1)!\,(n-\tau)!} \cdot \frac{\tau!\,(n-\tau)!}{n!} = \frac{(n-1)!}{n!} \cdot \frac{\tau!}{(\tau-1)!} = \frac{\tau}{n}. \qquad \square$$